COMPUTER NETWORK

LAB EXPERIMENTS – 1 to 16

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exp no – 1:

```
student@fedora:~$ arp -a
_gateway (10.0.2.2) at 52:55:0a:00:02:02 [ether] on enp0s3
```

```
student@fedora:~$ hostname
fedora
```

```
C:\Users\DELL>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : HDC0124069
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Ethernet Connection (17) I219-LM
   Physical Address. . . . . . . . . : 20-88-10-86-C6-AA
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-11
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::2709:30a3:dc42:768d%17(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 873070631
   DHCPv6 Client DUID. . . . . . . . : 00-01-01-00-31-2D-92-C3-20-88-10-86-C6-AA
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 4E-82-A9-77-FF-BB
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 42-82-A9-77-FF-BB
```

```
Wireless LAN adapter WiFi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
   Physical Address. . . . . . . . . : 4C-82-A9-77-FF-BB
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   IPv4 Address. . . . . . . . . . . : 172.16.75.9(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Default Gateway . . . . . . . . . : 172.16.72.1
   DNS Servers . . . . . . . . . . . : 172.16.72.1
                                       8.8.8.8
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

exp no – 1:

```
C:\Users\DELL>nbstat -a
'nbstat' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\DELL>nbtstat -a

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.
```

```
C:\Users\DELL>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49672        HDC0124069:49673       ESTABLISHED
  TCP    127.0.0.1:49673        HDC0124069:49672       ESTABLISHED
  TCP    127.0.0.1:49674        HDC0124069:49675       ESTABLISHED
  TCP    127.0.0.1:49675        HDC0124069:49674       ESTABLISHED
  TCP    172.16.75.9:55777      maa05s20-in-f10:https  ESTABLISHED
  TCP    172.16.75.9:56085      4.213.25.241:https     ESTABLISHED
```

```
student@fedora:~$ netstat -r
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
default         _gateway        0.0.0.0         UG        0 0          0 enp0s3
10.0.2.0        0.0.0.0         255.255.255.0   U         0 0          0 enp0s3
```

```
C:\Users\DELL>nslookup
Default Server:  UnKnown
Address:  172.16.72.1
```

```
PS C:\Users\DELL> pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list     Loose source route along host-list.
    -h maximum_hops  Maximum number of hops to search for target.
    -i address       Use the specified source address.
    -n               Do not resolve addresses to hostnames.
    -p period        Wait period milliseconds between pings.
    -q num_queries   Number of queries per hop.
    -w timeout       Wait timeout milliseconds for each reply.
    -4               Force using IPv4.
    -6               Force using IPv6.
```

exp no – 1:

```
PS C:\Users\DELL> ping

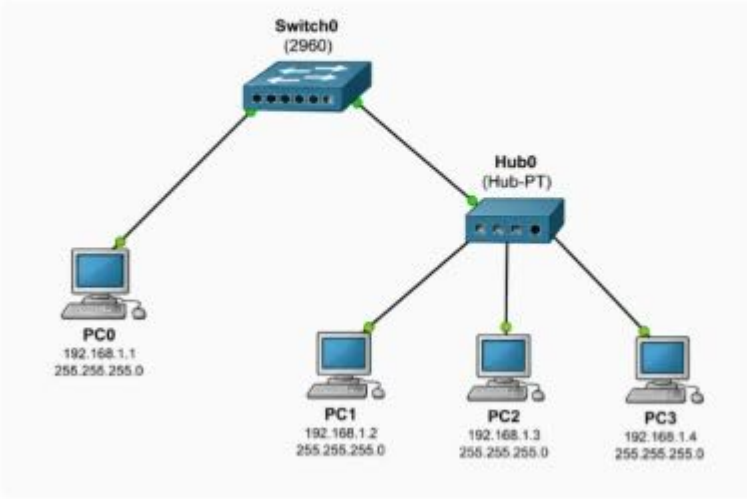
Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

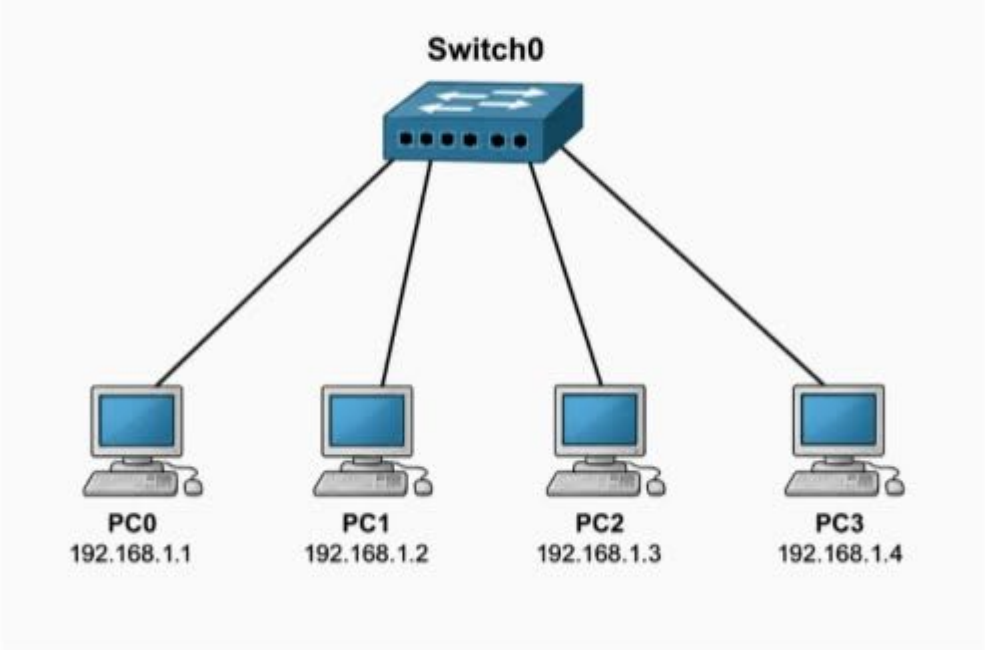
Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p             Ping a Hyper-V Network Virtualization provider address.
    -4             Force using IPv4.
    -6             Force using IPv6.
```

```
student@fedora:~$ route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         _gateway        0.0.0.0         UG    100    0        0 enp0s3
10.0.2.0        0.0.0.0         255.255.255.0   U     100    0        0 enp0s3
```

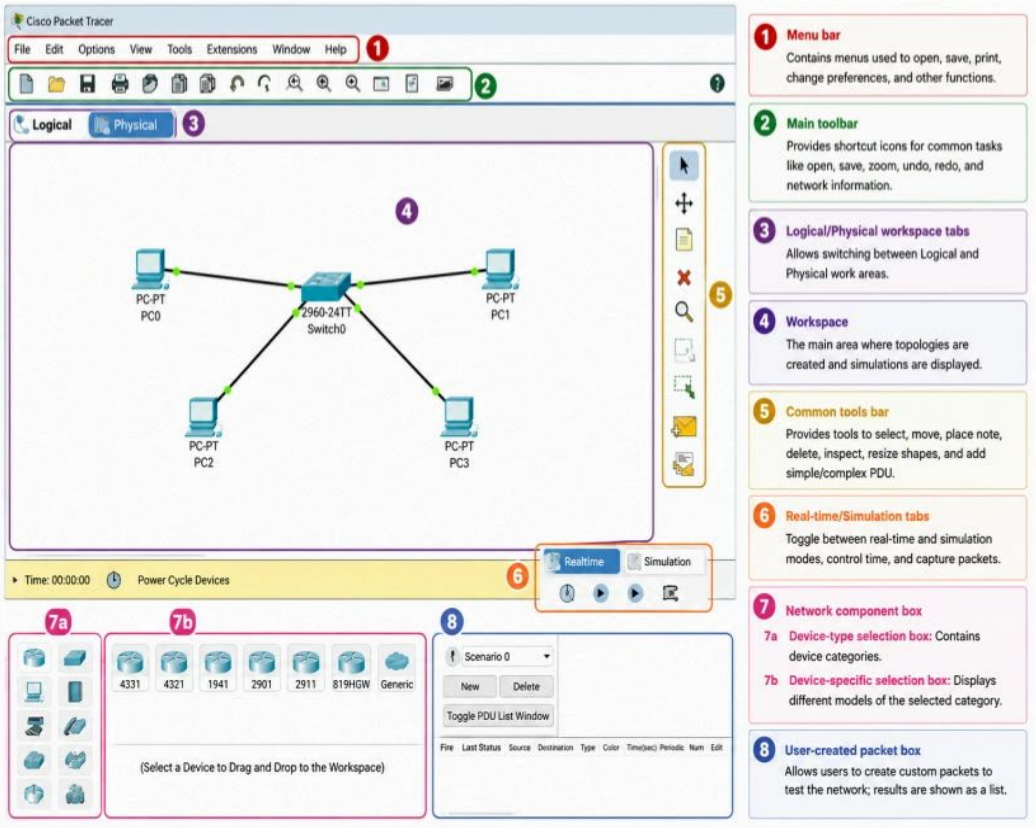
Exp no – 3:

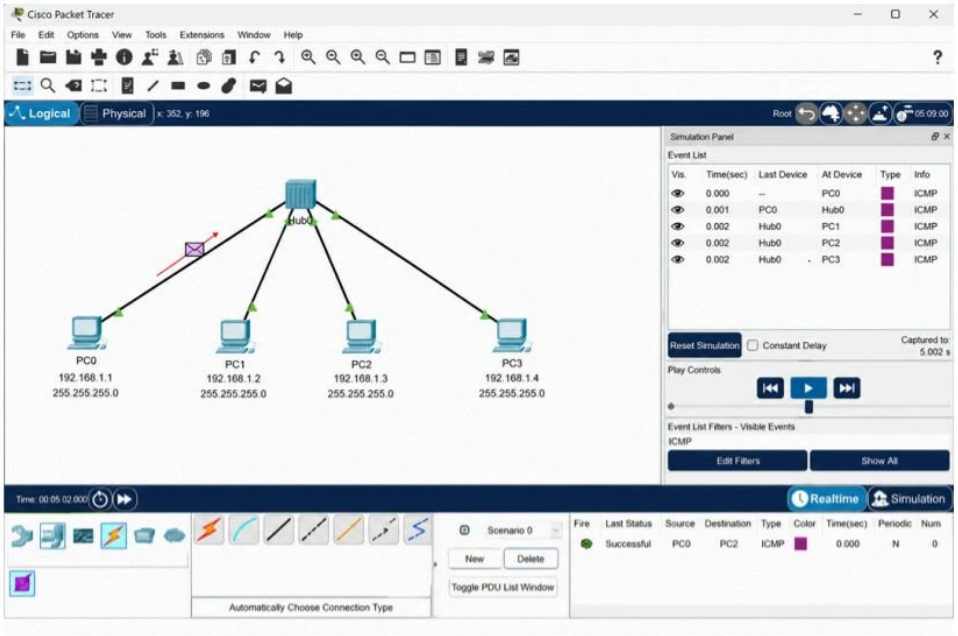
Network Topology



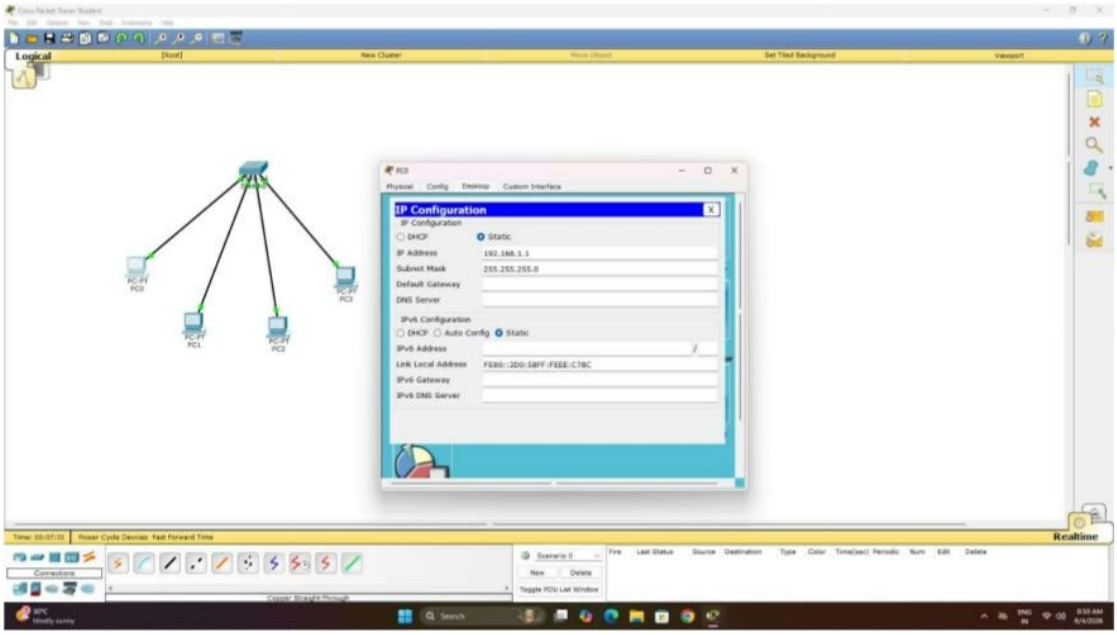


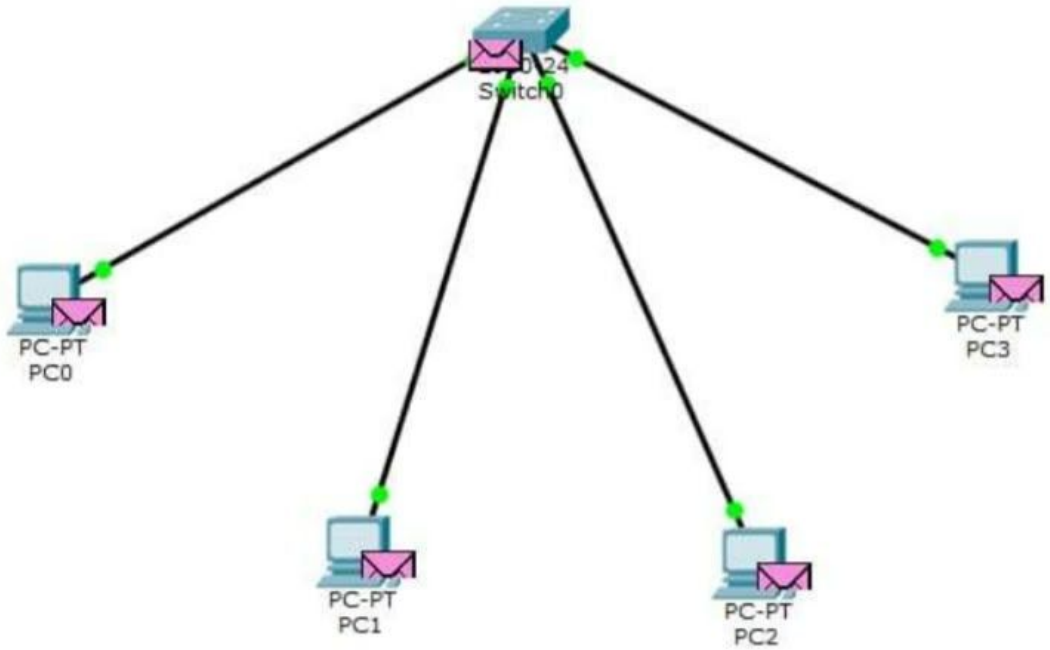
Exp no – 3:


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical
PC-PT PC0
2960-24TT Switch0
PC-PT PC1
PC-PT PC2
PC-PT PC3
Time: 00:00:00 Power Cycle Devices
Realtime Simulation
4331 4321 1941 2901 2911 819HGW Generic
(Select a Device to Drag and Drop to the Workspace)
Scenario 0 New Delete Toggle PDU List Window
1 Menu bar
Contains menus used to open, save, print, change preferences, and other functions.
2 Main toolbar
Provides shortcut icons for common tasks like open, save, zoom, undo, redo, and network information.
3 Logical/Physical workspace tabs
Allows switching between Logical and Physical work areas.
4 Workspace
The main area where topologies are created and simulations are displayed.
5 Common tools bar
Provides tools to select, move, place note, delete, inspect, resize shapes, and add simple/complex PDU.
6 Real-time/Simulation tabs
Toggle between real-time and simulation modes, control time, and capture packets.
7 Network component box
7a Device-type selection box: Contains device categories.
7b Device-specific selection box: Displays different models of the selected category.
8 User-created packet box
Allows users to create custom packets to test the network; results are shown as a list.

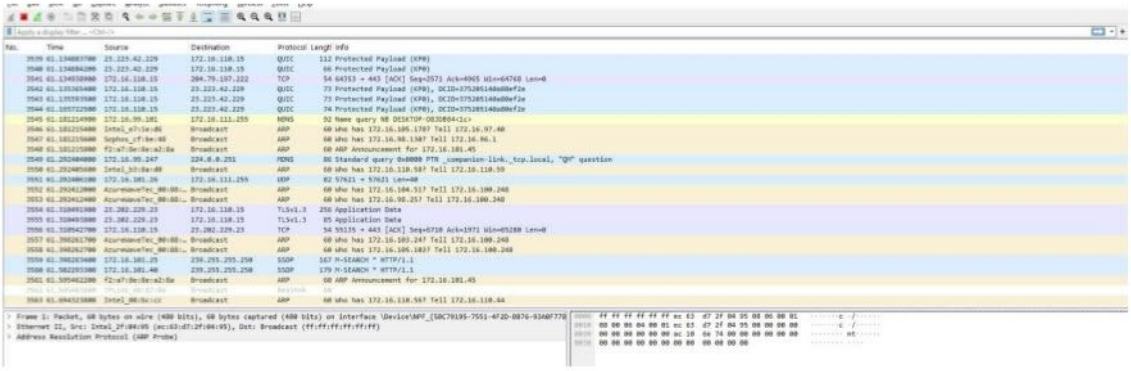

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical
Hub0
PC0 192.168.1.1 255.255.255.0
PC1 192.168.1.2 255.255.255.0
PC2 192.168.1.3 255.255.255.0
PC3 192.168.1.4 255.255.255.0
Simulation Panel
Event List
Vis. Time(sec) Last Device At Device Type Info
0.000 -- PC0 ICMP
0.001 PC0 Hub0 ICMP
0.002 Hub0 PC1 ICMP
0.002 Hub0 PC2 ICMP
0.002 Hub0 PC3 ICMP
Reset Simulation Constant Delay
Play Controls
Event List Filters - Visible Events
ICMP
Edit Filters Show All
Realtime Simulation
Scenario 0 New Delete Toggle PDU List Window
Automatically Choose Connection Type
Fire Last Status Source Destination Type Color Time(sec) Periodic Num
Successful PC0 PC2 ICMP 0.000 N 0

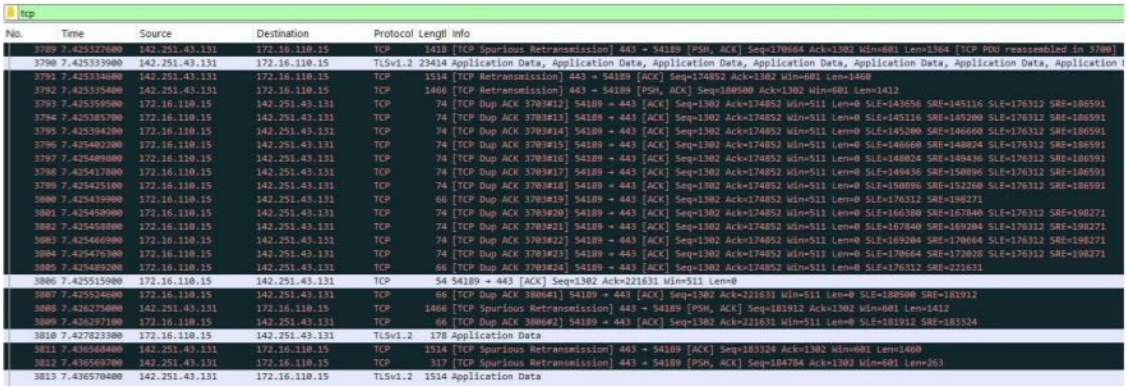
Exp no - 4:


Logical
IP Configuration
DHCP
Static
IP Address
192.168.1.1
Subnet Mask
255.255.255.0
Default Gateway
DNS Server
IPv6 Configuration
IPv6 Address
Link Local Address
IPv6 Gateway
IPv6 DNS Server
Realtime


Switch0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3

Exp no – 5:



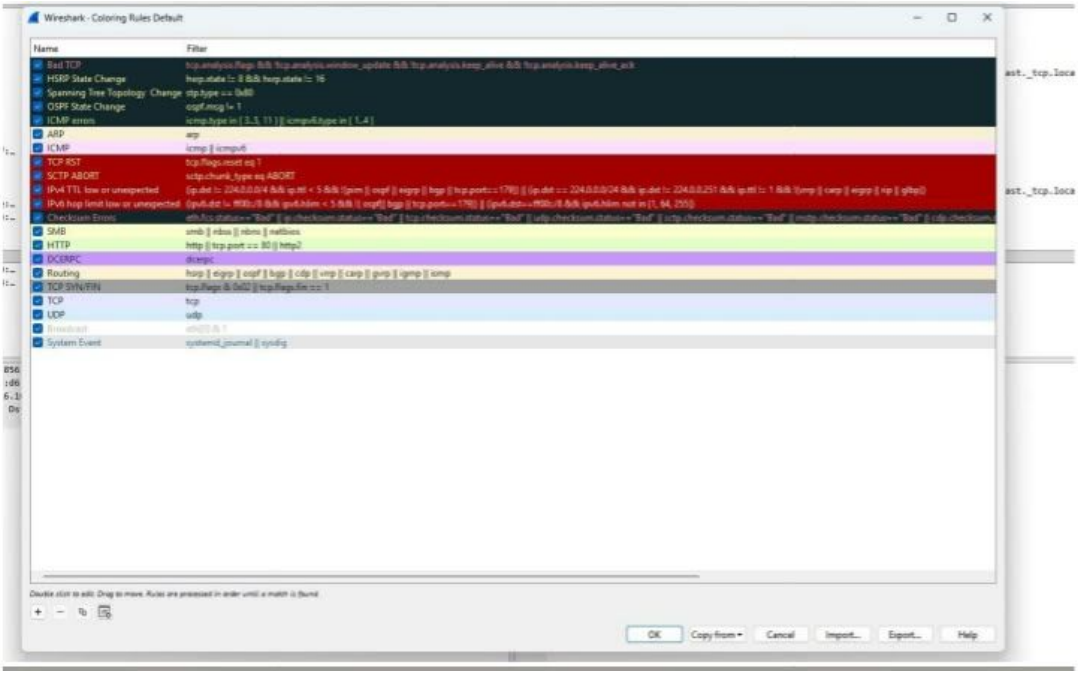


tcp

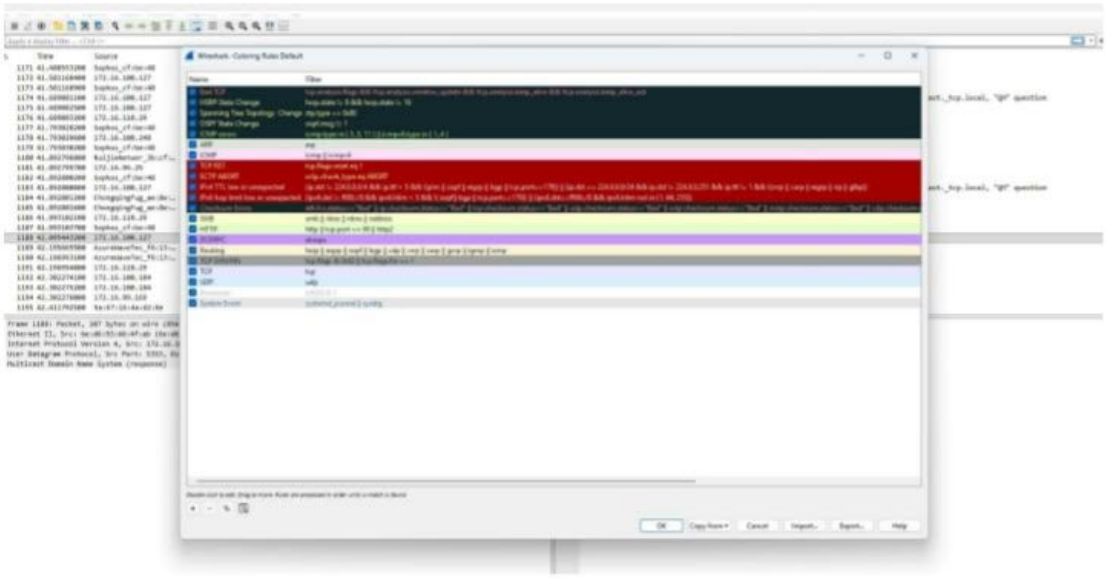
No.	Time	Source	Destination	Protocol	Length	Info
3789	7.425327600	142.251.43.131	172.16.110.15	TCP	1418	[TCP Spurious Retransmission] 443 → 54189 [PSH, ACK] Seq=170864 Ack=1302 Win=601 Len=1364 [TCP PDU reassembled in 3790]
3790	7.425333900	142.251.43.131	172.16.110.15	TLSv1.2	23414	Application Data, Application Data, Application Data, Application Data, Application Data, Application Data, Application
3791	7.425334600	142.251.43.131	172.16.110.15	TCP	1514	[TCP Retransmission] 443 → 54189 [ACK] Seq=174852 Ack=1302 Win=601 Len=1460
3792	7.425335400	142.251.43.131	172.16.110.15	TCP	1466	[TCP Retransmission] 443 → 54189 [PSH, ACK] Seq=180500 Ack=1302 Win=601 Len=1412
3793	7.425359500	172.16.110.15	142.251.43.131	TCP	74	[TCP Dup ACK 3703#12] 54189 → 443 [ACK] Seq=1302 Ack=174852 Win=511 Len=0 SLE=143656 SRE=145116 SLE=176312 SRE=186591
3794	7.425385700	172.16.110.15	142.251.43.131	TCP	74	[TCP Dup ACK 3703#13] 54189 → 443 [ACK] Seq=1302 Ack=174852 Win=511 Len=0 SLE=145116 SRE=145200 SLE=176312 SRE=186591
3795	7.425394200	172.16.110.15	142.251.43.131	TCP	74	[TCP Dup ACK 3703#14] 54189 → 443 [ACK] Seq=1302 Ack=174852 Win=511 Len=0 SLE=145200 SRE=146660 SLE=176312 SRE=186591
3796	7.425402200	172.16.110.15	142.251.43.131	TCP	74	[TCP Dup ACK 3703#15] 54189 → 443 [ACK] Seq=1302 Ack=174852 Win=511 Len=0 SLE=146660 SRE=148024 SLE=176312 SRE=186591
3797	7.425409800	172.16.110.15	142.251.43.131	TCP	74	[TCP Dup ACK 3703#16] 54189 → 443 [ACK] Seq=1302 Ack=174852 Win=511 Len=0 SLE=148024 SRE=149436 SLE=176312 SRE=186591
3798	7.425417800	172.16.110.15	142.251.43.131	TCP	74	[TCP Dup ACK 3703#17] 54189 → 443 [ACK] Seq=1302 Ack=174852 Win=511 Len=0 SLE=149436 SRE=150896 SLE=176312 SRE=186591
3799	7.425425100	172.16.110.15	142.251.43.131	TCP	74	[TCP Dup ACK 3703#18] 54189 → 443 [ACK] Seq=1302 Ack=174852 Win=511 Len=0 SLE=150896 SRE=152260 SLE=176312 SRE=186591
3800	7.425439900	172.16.110.15	142.251.43.131	TCP	66	[TCP Dup ACK 3703#19] 54189 → 443 [ACK] Seq=1302 Ack=174852 Win=511 Len=0 SLE=176312 SRE=198271
3801	7.425450900	172.16.110.15	142.251.43.131	TCP	74	[TCP Dup ACK 3703#20] 54189 → 443 [ACK] Seq=1302 Ack=174852 Win=511 Len=0 SLE=166380 SRE=167840 SLE=176312 SRE=198271
3802	7.425458800	172.16.110.15	142.251.43.131	TCP	74	[TCP Dup ACK 3703#21] 54189 → 443 [ACK] Seq=1302 Ack=174852 Win=511 Len=0 SLE=167840 SRE=169204 SLE=176312 SRE=198271
3803	7.425466900	172.16.110.15	142.251.43.131	TCP	74	[TCP Dup ACK 3703#22] 54189 → 443 [ACK] Seq=1302 Ack=174852 Win=511 Len=0 SLE=169204 SRE=170664 SLE=176312 SRE=198271
3804	7.425476300	172.16.110.15	142.251.43.131	TCP	74	[TCP Dup ACK 3703#23] 54189 → 443 [ACK] Seq=1302 Ack=174852 Win=511 Len=0 SLE=170664 SRE=172028 SLE=176312 SRE=198271
3805	7.425489200	172.16.110.15	142.251.43.131	TCP	66	[TCP Dup ACK 3703#24] 54189 → 443 [ACK] Seq=1302 Ack=174852 Win=511 Len=0 SLE=176312 SRE=221631
3806	7.425515900	172.16.110.15	142.251.43.131	TCP	54	54189 → 443 [ACK] Seq=1302 Ack=221631 Win=511 Len=0
3807	7.425524600	172.16.110.15	142.251.43.131	TCP	66	[TCP Dup ACK 3806#1] 54189 → 443 [ACK] Seq=1302 Ack=221631 Win=511 Len=0 SLE=180500 SRE=181912
3808	7.426275000	142.251.43.131	172.16.110.15	TCP	1466	[TCP Spurious Retransmission] 443 → 54189 [PSH, ACK] Seq=181912 Ack=1302 Win=601 Len=1412
3809	7.426297100	172.16.110.15	142.251.43.131	TCP	66	[TCP Dup ACK 3806#2] 54189 → 443 [ACK] Seq=1302 Ack=221631 Win=511 Len=0 SLE=181912 SRE=183324
3810	7.427823300	172.16.110.15	142.251.43.131	TLSv1.2	178	Application Data
3811	7.436568400	142.251.43.131	172.16.110.15	TCP	1514	[TCP Spurious Retransmission] 443 → 54189 [ACK] Seq=183324 Ack=1302 Win=601 Len=1460
3812	7.436569700	142.251.43.131	172.16.110.15	TCP	317	[TCP Spurious Retransmission] 443 → 54189 [PSH, ACK] Seq=184784 Ack=1302 Win=601 Len=263
3813	7.436570400	142.251.43.131	172.16.110.15	TLSv1.2	1514	Application Data

Exp no – 5:


Time
142.251.222.165
172.16.110.15
142.251.221.142
0.036204600 443 Application Data 53258
0.036205900 443 Application Data 53258
0.036206600 443 Application Data 53258
0.036207100 443 Application Data 53258
0.036213200 443 Application Data 53258
0.036216100 443 Application Data, Application Data 53258
0.036219100 443 Application Data 53258
0.036219500 443 Application Data 53258
0.036265300 443 53258 → 443 [ACK] Seq=1 Ack=9377 Win=1023 Len=0 53258
0.036627900 443 Application Data 53258
0.042442900 443 Application Data 53258
0.061638300 443 443 → 53258 [ACK] Seq=9377 Ack=40 Win=6155 Len=0 53258
0.061638700 443 443 → 53258 [ACK] Seq=9377 Ack=1460 Win=6178 Len=0 53258
0.061638900 443 443 → 53258 [ACK] Seq=9377 Ack=2880 Win=6200 Len=0 53258
0.061639100 443 443 → 53258 [ACK] Seq=9377 Ack=4300 Win=6223 Len=0 53258
0.061639300 443 443 → 53258 [ACK] Seq=9377 Ack=6336 Win=6255 Len=0 53258
0.091769300 50906 Application Data 443
0.091825900 50906 Application Data 443
0.114040900 50906 443 → 50906 [ACK] Seq=1 Ack=1461 Win=6606 Len=0 443
0.114041900 50906 443 → 50906 [ACK] Seq=1 Ack=3735 Win=6642 Len=0 443
0.114042300 50906 443 → 50906 [ACK] Seq=1 Ack=3860 Win=6642 Len=0 443
0.178539300 443 Application Data 53258
0.178750100 443 Application Data 53258
0.178751100 443 Application Data 53258
0.178773400 443 53258 → 443 [ACK] Seq=6336 Ack=9782 Win=1022 Len=0 53258
0.178997400 443 Application Data 53258
0.185103300 443 Application Data 53258


Wireshark · Coloring Rules Default
Name
Filter
Bad TCP
HSRP State Change
Spanning Tree Topology Change
OSPF State Change
ICMP errors
ARP
ICMP
TCP RST
SCTP ABORT
IPv4 TTL low or unexpected
IPv6 hop limit low or unexpected
Checksum Errors
SMB
HTTP
DCERPC
Routing
TCP SYN/FIN
TCP
UDP
Broadcast
System Event
OK
Copy from
Cancel
Import...
Export...
Help

Exp no – 5:



Exp no – 6:

```
Reply from google.com: time=2836.94 ms
Reply from google.com: time=931.80 ms
Reply from google.com: time=157.00 ms
Reply from google.com: time=65.24 ms
```

Exp no – 7:

```
PS C:\Windows\system32> cd "C:\Users\REC\AppData\Local\Programs\Python\Python314"
PS C:\Users\REC\AppData\Local\Programs\Python\Python314> python cn7.py
Listening on: 192.168.56.1

Packet Sniffer Started...


Packet 1
----------------------------------------
Destination MAC : 45:00:01:48:3D:E3
Source MAC      : 00:00:01:11:A1:1D
Protocol        : 0xc0a8
Packet Length   : 328 bytes

Packet 2
----------------------------------------
Destination MAC : 45:00:01:48:3D:E3
Source MAC      : 00:00:01:11:A1:1D
Protocol        : 0xc0a8
Packet Length   : 328 bytes

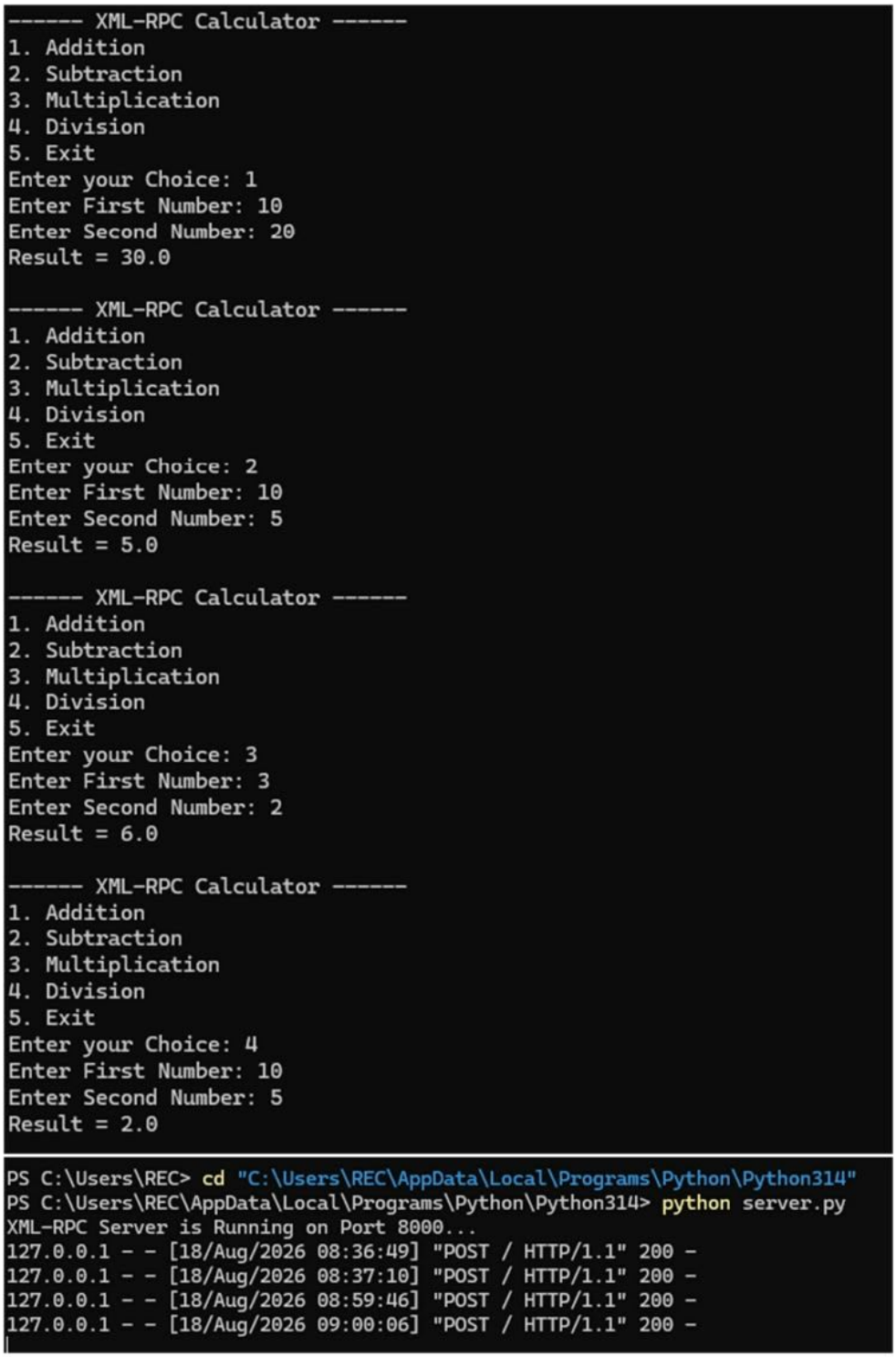
Packet 3
----------------------------------------
Destination MAC : 45:00:00:4F:3D:E4
Source MAC      : 00:00:01:11:A2:15
Protocol        : 0xc0a8
Packet Length   : 79 bytes

Packet 4
----------------------------------------
Destination MAC : 45:00:00:4F:3D:E4
Source MAC      : 00:00:01:11:A2:15
Protocol        : 0xc0a8
Packet Length   : 79 bytes

Packet 5
----------------------------------------
Destination MAC : 45:00:00:4F:3D:E5
Source MAC      : 00:00:01:11:A2:14
Protocol        : 0xc0a8
Packet Length   : 79 bytes

Packet Sniffer Stopped Successfully.
PS C:\Users\REC\AppData\Local\Programs\Python\Python314>
```

Exp no – 8:


------ XML-RPC Calculator ------
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your Choice: 1
Enter First Number: 10
Enter Second Number: 20
Result = 30.0
------ XML-RPC Calculator ------
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your Choice: 2
Enter First Number: 10
Enter Second Number: 5
Result = 5.0
------ XML-RPC Calculator ------
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your Choice: 3
Enter First Number: 3
Enter Second Number: 2
Result = 6.0
------ XML-RPC Calculator ------
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your Choice: 4
Enter First Number: 10
Enter Second Number: 5
Result = 2.0
PS C:\Users\REC> cd "C:\Users\REC\AppData\Local\Programs\Python\Python314"
PS C:\Users\REC\AppData\Local\Programs\Python\Python314> python server.py
XML-RPC Server is Running on Port 8000...
127.0.0.1 - - [18/Aug/2026 08:36:49] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [18/Aug/2026 08:37:10] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [18/Aug/2026 08:59:46] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [18/Aug/2026 09:00:06] "POST / HTTP/1.1" 200 -

`

Exp no – 9:

```
PS C:\Users\REC>  cd "C:\Users\REC\AppData\Local\Programs\Python\Python314"
PS C:\Users\REC\AppData\Local\Programs\Python\Python314> python client9.py
Connected Successfully
Available Commands
1. hello
2. time
3. date
4. exit

Enter Command: hello
Server: Hello from Server!

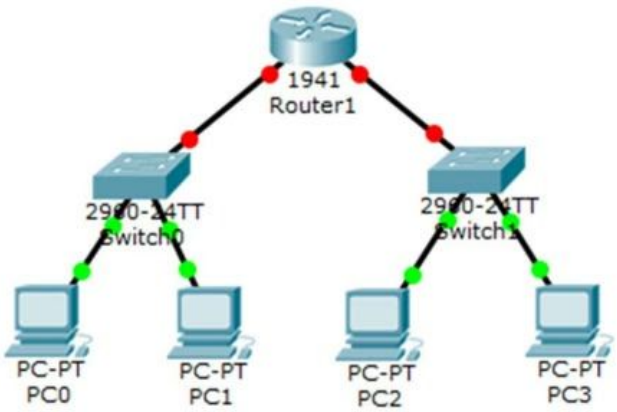
Enter Command: time
Server: 08:53:08

Enter Command: date
Server: 18-08-2026

Enter Command: exit
Server: Connection Closed
PS C:\Users\REC\AppData\Local\Programs\Python\Python314>
```

```
PS C:\Users\REC>  cd "C:\Users\REC\AppData\Local\Programs\Python\Python314"
PS C:\Users\REC\AppData\Local\Programs\Python\Python314> python server9.py
Server is listening on 127.0.0.1 5000
Connected by: ('127.0.0.1', 50246)
Client Command: hello
Client Command: time
Client Command: date
Client Command: exit
PS C:\Users\REC\AppData\Local\Programs\Python\Python314>
```

Exp no – 10:



FastEthernet0

Port Status: On

Bandwidth: 100 Mbps / 10 Mbps / Auto

Duplex: Half Duplex / Full Duplex / Auto

MAC Address: 0060.2FB2.B128

IP Configuration

- DHCP
- Static

IP Address: 192.168.1.10

Subnet Mask: 255.255.255.0

IPv6 Configuration

- DHCP
- Auto Config
- Static

IPv6 Address: /

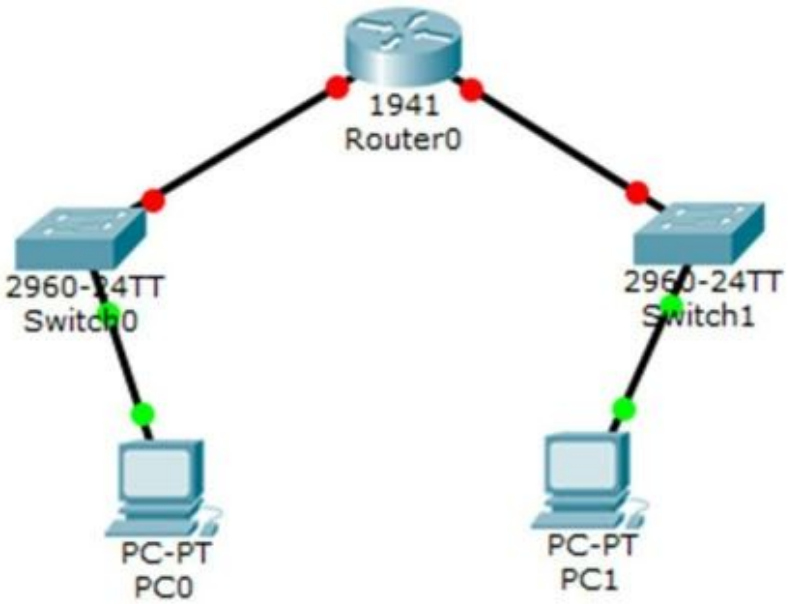
Link Local Address: FE80::260:2FFF:FEB2:B128

Exp no – 11:

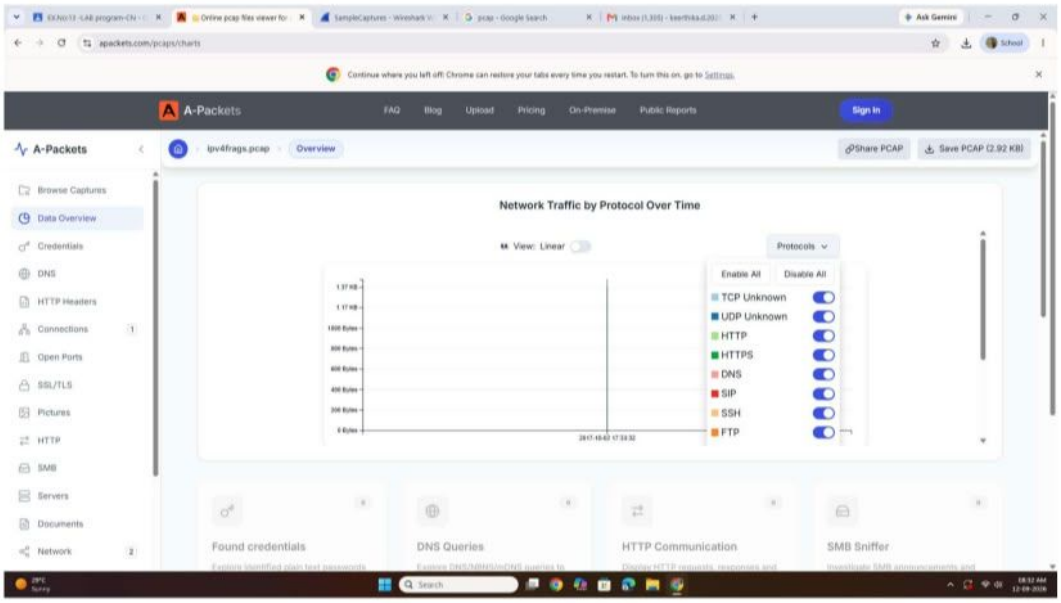
```
PS C:\WINDOWS\system32> cd "C:\Users\REC\AppData\Local\Programs\Python\Python314"
PS C:\Users\REC\AppData\Local\Programs\Python\Python314> & "C:\Users\REC\AppData\Local\Programs\P
ython\Python314\python.exe" "C:\Users\REC\AppData\Local\Programs\Python\Python314\exp10s.py"
Server is waiting for connection...
Client connected: ('127.0.0.1', 50506)
Message from client: Hello Server!
PS C:\Users\REC\AppData\Local\Programs\Python\Python314>
```

```
PS C:\WINDOWS\system32> cd "C:\Users\REC\AppData\Local\Programs\Python\Python314"
PS C:\Users\REC\AppData\Local\Programs\Python\Python314> & "C:\Users\REC\AppData\Local\Programs\P
ython\Python314\python.exe" "C:\Users\REC\AppData\Local\Programs\Python\Python314\exp10c.py"
Message from server: Hello Client! Message received successfully
PS C:\Users\REC\AppData\Local\Programs\Python\Python314>
```

Exp no – 12:


1941
Router0
2960-24TT
Switch0
2960-24TT
Switch1
PC-PT
PC0
PC-PT
PC1

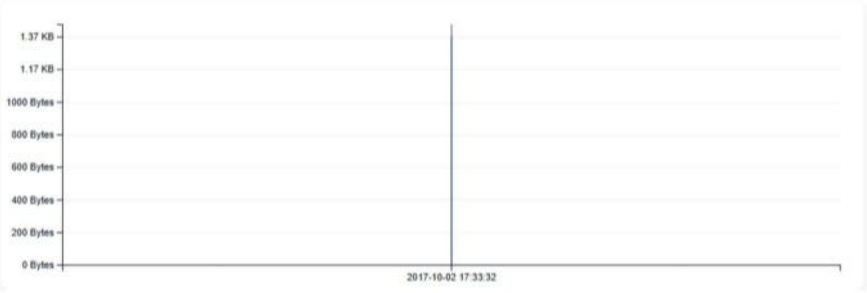
Exp no – 13:



Network Traffic by Protocol Over Time

View: Linear

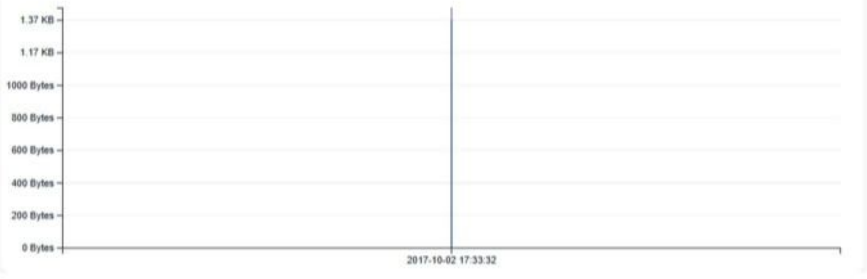
Protocols



Network Traffic by Protocol Over Time

View: Linear

Protocols



Exp no – 15:

```
Enter "help" below or click "Help" above for more information.
>
===== RESTART: C:/Users/REC/AppData/Local/Programs/Python/Python314/15cn.py ====
230-Welcome to mirror.as35701.net.

 The server is located in Brussels, Belgium.

 Server connected with gigabit ethernet to the internet.

 The server maintains software archive accessible via ftp, http, https and rsync.

 ftp.be.debian.org is an alias for this host, but https will not work with that
 alias. If you want to use https use mirror.as35701.net.

 Contact: mirror-admin at as35701.net

230 Anonymous access granted, restrictions apply

[*] FTP Anonymous Login Succeeded.
Server: ftp.be.debian.org

Welcome Message:
220 ProFTPD Server (mirror.as35701.net) [::ffff:195.234.45.114]

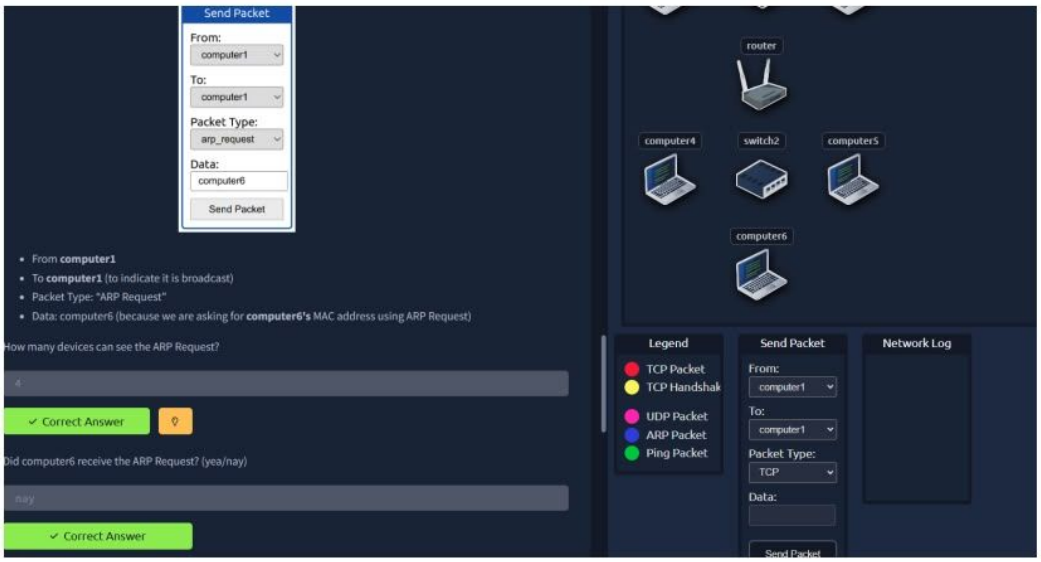
Directory Listing:
drwxr-xr-x   9 ftp      ftp          4096 Oct  6 02:53 debian
drwxr-xr-x   5 ftp      ftp           105 Sep 13 00:27 debian-cd
drwxr-xr-x   7 ftp      ftp          4096 Oct  5 18:28 debian-security
drwxr-xr-x   5 ftp      ftp          4096 Oct 13  2006 ftp.irc.org
-rw-r--r--   1 ftp      ftp           717 Apr 15  2025 HEADER.html
drwxr-xr-x   5 ftp      ftp          4096 Oct  5 23:24 mint
drwxr-xr-x   6 ftp      ftp            60 Apr 27 10:28 mint-iso
lrwxrwxrwx   1 ftp      ftp            33 Apr 29  2021 pub -> /var/www/html/www.kernel.org/pub/
drwxr-xr-x   7 ftp      ftp          4096 Oct  6 00:50 ubuntu
drwxr-xr-x  31 ftp      ftp          4096 Oct  5 18:49 ubuntu-cdimage
drwxr-xr-x  31 ftp      ftp          4096 Oct  5 19:17 ubuntu-cloudimages
drwxr-xr-x   7 ftp      ftp          4096 Oct  6 01:29 ubuntu-ports
drwxr-xr-x  17 ftp      ftp          4096 Oct  6 01:22 ubuntu-releases
drwxr-xr-x  26 ftp      ftp           315 Oct  5 23:06 video.fosdem.org
-rw-r--r--   1 ftp      ftp           390 Jul  9  2021 welcome.msg
drwxr-xr-x   4 ftp      ftp          4096 Jun 14  2023 www.kernel.org
>
```

Exp no – 16:

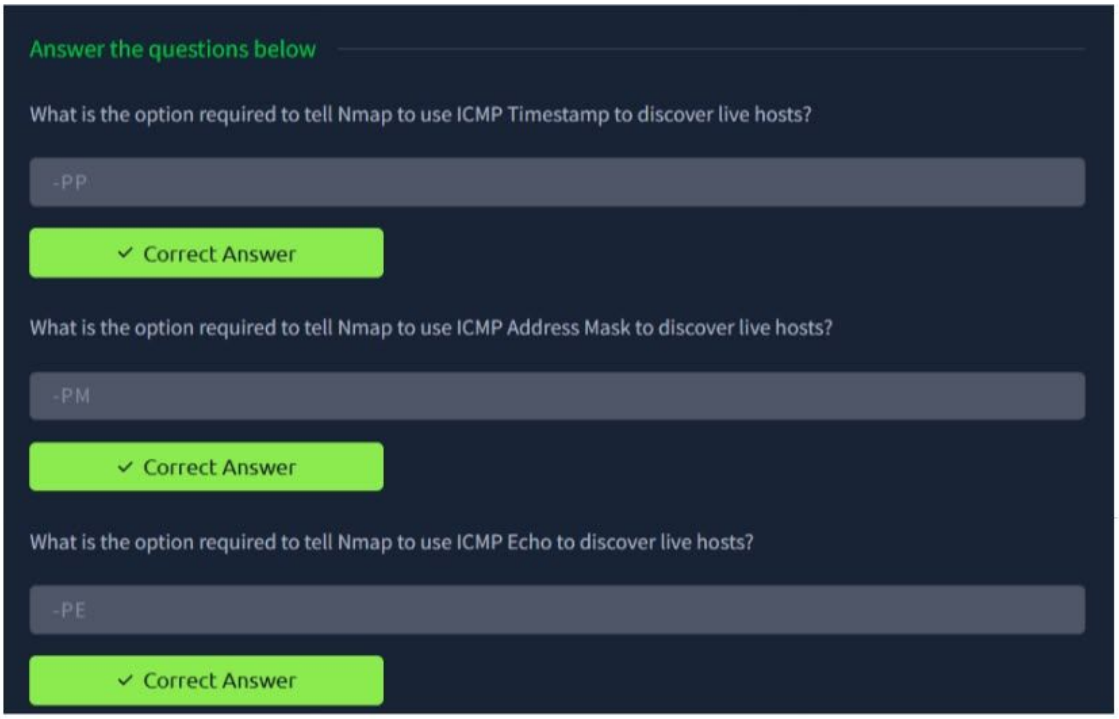

Nmap Live Host Discovery
Learn how to use Nmap to discover live hosts using ARP, ICMP, and TCP/UDP ping scan.
75 min
297,362
Share your achievement
Start AttackBox
Save Room
5423 Recommend
Connectivity
Options
Room completed: 100%

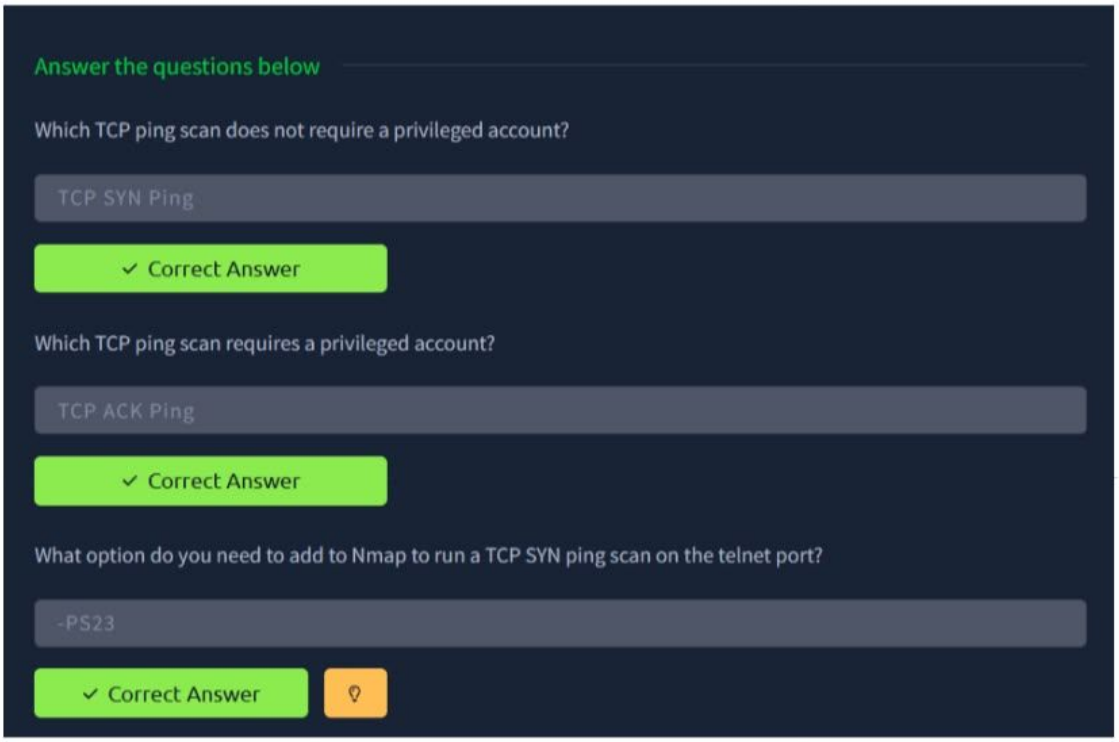
Exp no – 16:


Task 1 Introduction
Task 2 Subnetworks
Task 3 Understanding Hosts Discovery through TCP/IP Layer
Task 4 Enumerating Targets
Task 5 Nmap Host Discovery Using ARP
Task 6 Nmap Host Discovery Using ICMP
Task 7 Nmap Host Discovery Using TCP and UDP
Task 8 Using Reverse-DNS Lookup
Task 9 Summary


Send Packet
From:
computer1
To:
computer1
Packet Type:
arp_request
Data:
computer6
Send Packet
From computer1
To computer1 (to indicate it is broadcast)
Packet Type: "ARP Request"
Data: computer6 (because we are asking for computer6's MAC address using ARP Request)
How many devices can see the ARP Request?
4
Correct Answer
Did computer6 receive the ARP Request? (yea/nay)
nay
Correct Answer
router
computer4
switch2
computer5
computer6
Legend
TCP Packet
TCP Handshak
UDP Packet
ARP Packet
Ping Packet
Send Packet
From:
computer1
To:
computer1
Packet Type:
TCP
Data:
Network Log

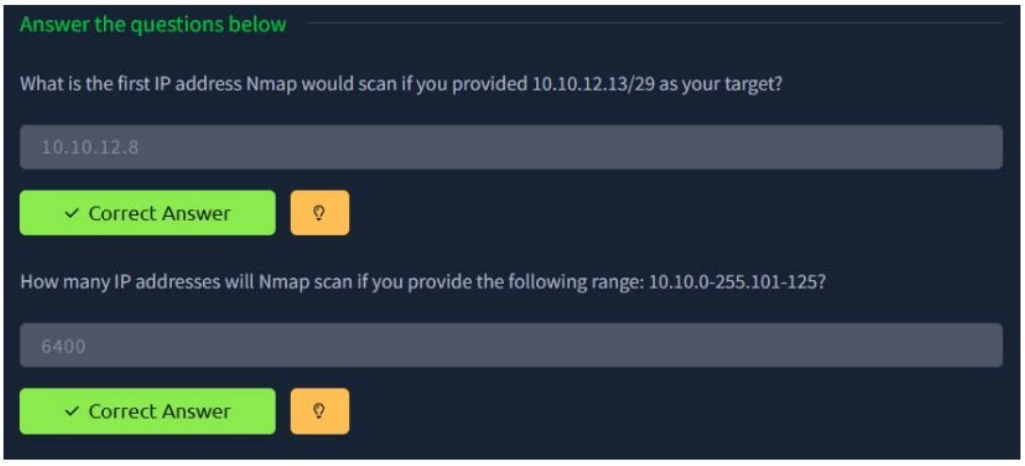
Exp no – 16:


Answer the questions below
What is the option required to tell Nmap to use ICMP Timestamp to discover live hosts?
-PP
✓ Correct Answer
What is the option required to tell Nmap to use ICMP Address Mask to discover live hosts?
-PM
✓ Correct Answer
What is the option required to tell Nmap to use ICMP Echo to discover live hosts?
-PE
✓ Correct Answer


Answer the questions below
Which TCP ping scan does not require a privileged account?
TCP SYN Ping
✓ Correct Answer
Which TCP ping scan requires a privileged account?
TCP ACK Ping
✓ Correct Answer
What option do you need to add to Nmap to run a TCP SYN ping scan on the telnet port?
-PS23
✓ Correct Answer

Exp no – 16:


Send a packet with the following:
From **computer2**
To **computer5**
Packet Type: "**Ping Request**"
What is the name of the first device that responded to the first ARP Request?
router
✓ Correct Answer
What is the name of the first device that responded to the second ARP Request?
computer5
✓ Correct Answer
Send another Ping Request. Did it require new ARP Requests? (yea/nay)
nay
✓ Correct Answer


Answer the questions below
What is the first IP address Nmap would scan if you provided 10.10.12.13/29 as your target?
10.10.12.8
✓ Correct Answer
How many IP addresses will Nmap scan if you provide the following range: 10.10.0-255.101-125?
6400
✓ Correct Answer


Answer the questions below
How many hosts are found alive after scanning the CONNECTION_IP/24?
1
✓ Correct Answer